





# Data Modelling Fundamentals and Entity- Relationship (ER) Model

Building the Blueprint of a Database

# Learning Outcomes

- By the end of our session, we will be able to:
- Understand the purpose of data Modelling.
- Define Entity, Attribute, and Relationship.
- Identify different types of entities and attributes.
- Understand keys in ER Modelling.
- Draw simple ER diagrams.
- Apply ER Modelling to real-world problems.



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# Why Do We Need Data Modelling?

- Before constructing a building, an architect prepares a blueprint.
- Similarly, before creating a database, a designer prepares a data model.

## Definition

Data Modelling is the process of identifying, organizing, and representing data and the relationships among data before implementing the database.

- Objectives
- Understand business requirements
- Organize information logically
- Reduce redundancy
- Improve consistency
- Simplify database design
- Facilitate communication between users and developers
- Example: For creating University Database, instead of immediately creating tables, we first identify:
  - Students
  - Courses
  - Faculty
  - Departments
  - Relationships among them



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# Levels of Data Modelling

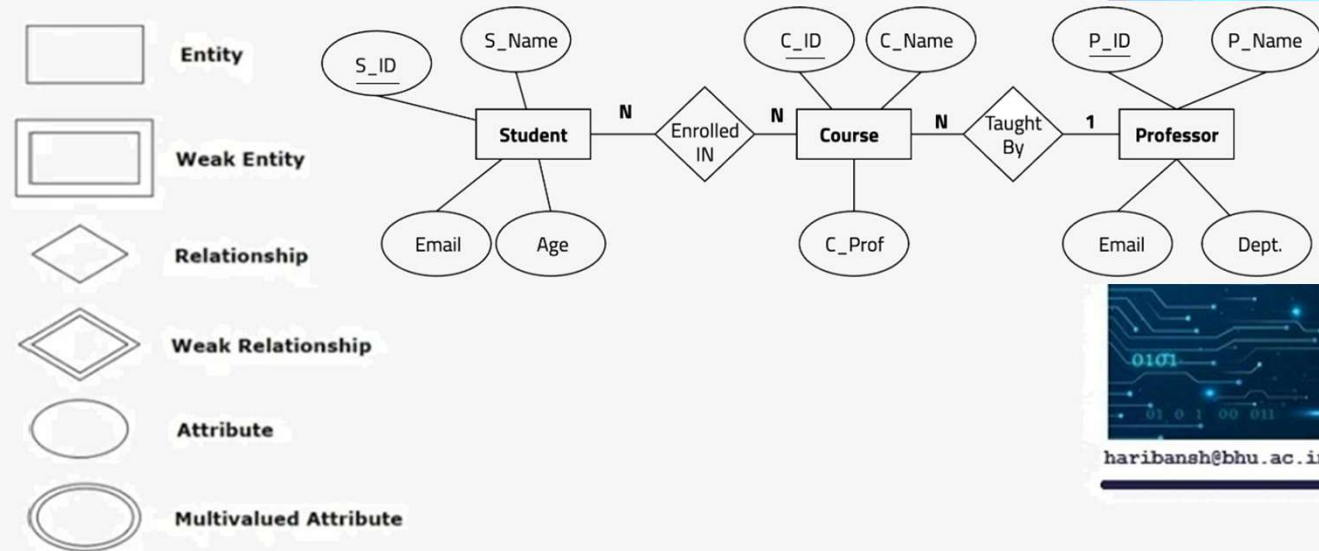
- Database design is usually performed in three stages.
- *1. Conceptual Data Model*
  - Focuses on
  - Entities
  - Relationships
  - Business rules
  - Tool used: ER Diagram
- *2. Logical Data Model*
  - Focuses on
  - Tables
  - Primary Keys
  - Foreign Keys
  - Constraints
  - Independent of DBMS implementation.
- *3. Physical Data Model*
  - Focuses on
  - Storage
  - Indexes
  - File organization
  - Performance optimization
  - Dependent on the chosen DBMS.



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# What is the Entity-Relationship (ER) Model?

- Definition
- The Entity-Relationship (ER) Model is a high-level conceptual data model used to represent data, relationships, and constraints of a database.
- It was proposed by Peter Chen (1976).
- Purpose
  - Visualize database design
  - Capture user requirements
  - Improve communication
  - Reduce design errors
- ER Diagram Symbols
  - Symbol Meaning
  - Rectangle Entity
  - Oval Attribute
  - Diamond Relationship
  - Line Connection



# Entity

- Definition
- An Entity is a real-world object, person, place, event, or concept about which information is stored.
- Characteristics
  - Has independent existence
  - Can be uniquely identified
  - Has attributes
- Examples
  - University
  - Student
  - Faculty
  - Department
  - Course
  - Bank
  - Customer
  - Account
  - Branch
  - Hospital
  - Patient
  - Doctor
  - Medicine



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# Entity Set and Types of Entities

- Entity Set
- A collection of similar entities.
- Example
  - *Student Entity Set*
  - Rahul
  - Anita
  - Riya
  - Vikram
- Types of Entities
- Strong Entity: Exists independently.
  - *Example*
  - Student
  - Department
  - Employee
  - Represented by: Single Rectangle
- Weak Entity: Depends on another entity for its existence.
  - *Example*
  - Dependent (Employee Insurance)
  - Order Item
  - Semester Registration
  - Represented by: Double Rectangle



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# Attributes

- Definition
- An Attribute is a property or characteristic of an entity.
- Example
  - *Student*
  - Roll Number
  - Name
  - Email
  - Age
  - Phone
- Representation
- Attributes are represented using Ovals.



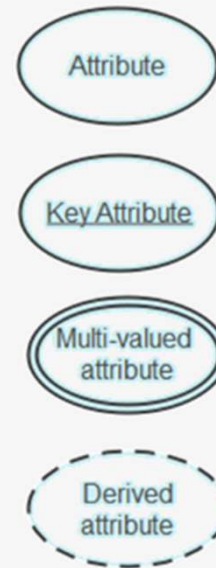
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# Types of Attributes

- **1. Simple Attribute**
- Cannot be divided further.
  - Example
  - Gender
  - Age
- **2. Composite Attribute**
- Can be divided into smaller parts.
  - Example
  - Address
    - House No
    - Street
    - City
    - State
    - PIN Code
- **3. Single-Valued Attribute**
- Only one value.
  - Example
  - Date of Birth
- **4. Multi-Valued Attribute**
- Multiple values.
- Example
  - Phone Numbers
  - Skills
  - Languages Known
- Represented using Double Oval.

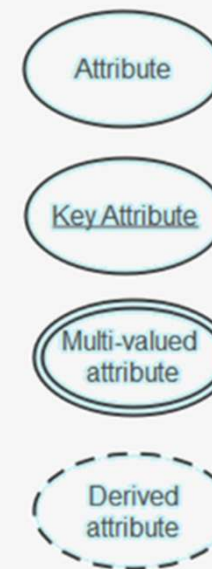


Composite Attribute



# More Types of Attributes

- *5. Derived Attribute*
- Computed from another attribute.
  - Example
    - Age
    - *Derived from Date of Birth*
- Represented by: Dashed Oval
- *6. Stored Attribute*
- Actually stored in database.
  - Example
  - Date of Birth



# Keys in ER Modelling

- Definition
  - A Key Attribute uniquely identifies each entity in an entity set.
    - Example
    - *Student*
- | Roll No |  | Name  |
|---------|--|-------|
| =====   |  |       |
| 101     |  | Rahul |
| 102     |  | Anita |
- Roll Number uniquely identifies each student.
  - Types
    - Candidate Key
    - Primary Key
    - Alternate Key
    - Composite Key
  - Note: Keys will be discussed in detail latter
  - Represented by: Underlined Attribute



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# Relationships

- Definition
- A Relationship is an association between two or more entities.
- Represented using a Diamond.
  - Examples
    - Student *Enrolls* Course
    - Faculty *Teaches* Course
    - Doctor *Treats* Patient
- *Importance*
- Relationships capture business rules.



# Degree of Relationship

- *Unary Relationship*
- Entity relates to itself.
- Example
  - Employee supervises Employee.

- *Binary Relationship*
- Between two entities.
- Example
  - Student enrolls in Course.

- *Ternary Relationship*
  - Among three entities.

- Example
  - Supplier supplies Part to Project.
- Comparison

| Degree  | Number of Entities |
|---------|--------------------|
| =====   |                    |
| Unary   | 1                  |
| Binary  | 2                  |
| Ternary | 3                  |



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# Cardinality Constraints

## Definition

Cardinality specifies how many entities can participate in a relationship.

- *Types*

- One-to-One (1:1)

One Person



One Passport

- One-to-Many (1:M)

One Department



Many Students

- Many-to-One (M:1)

Many Employees



One Department

- Many-to-Many (M:N)

Many Students



Many Courses



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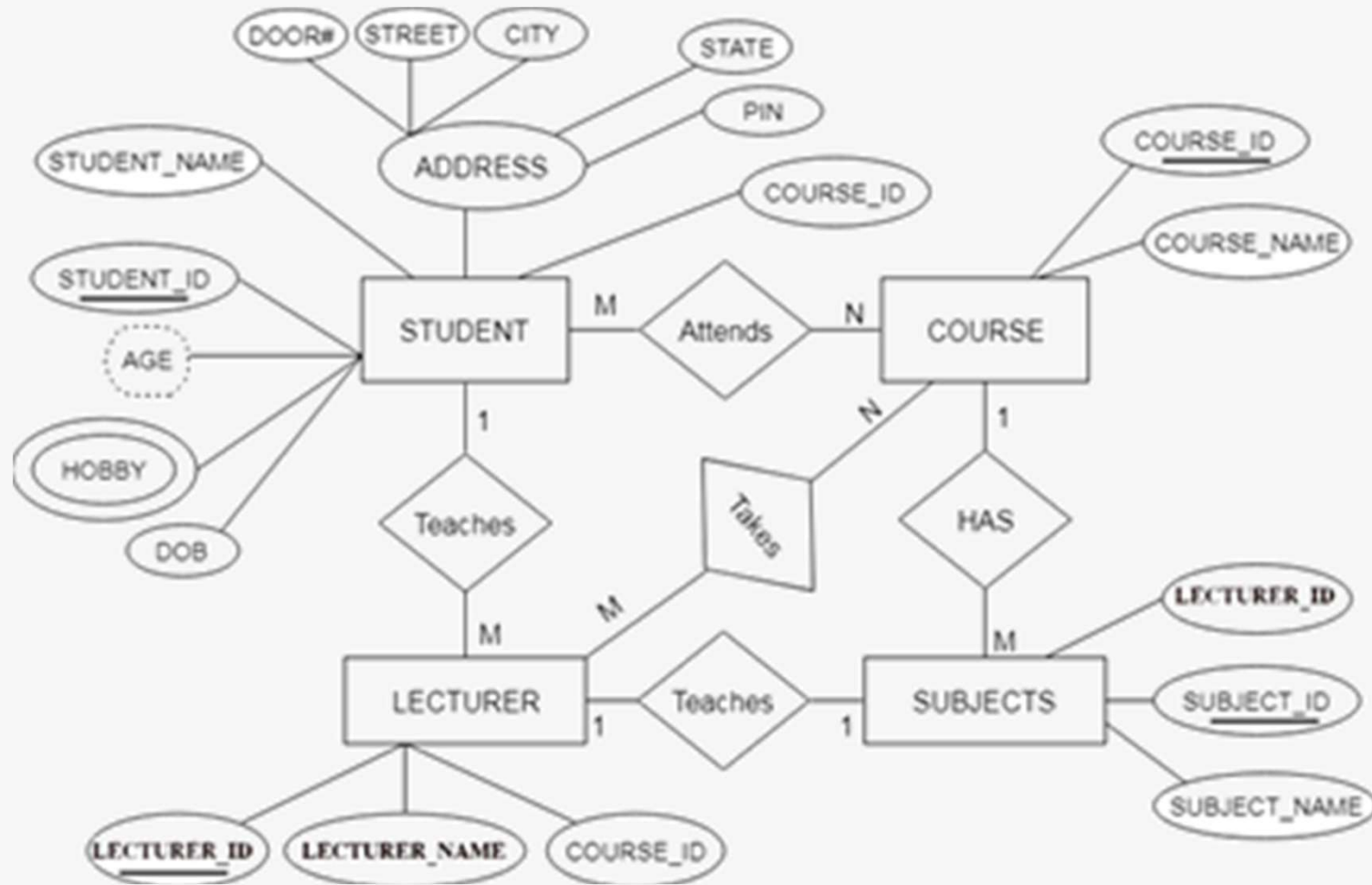


# Participation Constraints

- Participation specifies whether the participation of an entity in a relationship is mandatory or optional.
- *Total Participation*
  - Every entity must participate.
  - Example
  - Every Employee must belong to a Department.
  - Represented by
  - Double Line
- *Partial Participation*
  - Participation is optional.
  - Example
  - Some Faculty members may not supervise research scholars.
  - Represented by
  - Single Line



# ER Diagram Example



# Modern Perspective

- Although relational databases dominate enterprise systems,
- the idea of entities and relationships is still fundamental in modern databases.
- Examples
- *Relational Database*

Entity



Table

Entity



Collection  
Document

- *MongoDB*

- *Graph Database (Neo4j)*

Entities become Nodes

Relationships become Edges

- Important Note: Different databases store data differently, but data Modelling remains essential.



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# Summary

- Today we learned
- ✓ Data Modelling
- ✓ Levels of Data Modelling
- ✓ ER Model
- ✓ Entities
- ✓ Entity Sets
- ✓ Strong Entity
- ✓ Weak Entity
- ✓ Attributes
- ✓ Types of Attributes
- ✓ Keys
- ✓ Relationships
- ✓ Degree of Relationships
- ✓ Cardinality
- ✓ Participation Constraints
- ✓ Modern Perspective



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# Learning Reflections

- Define data Modelling. Why is it important?
- Differentiate between an entity and an entity set.
- Explain strong and weak entities with examples.
- What are the different types of attributes?
- Define relationship and relationship degree.
- Explain cardinality constraints with suitable examples.
- How are ER concepts reflected in NoSQL and graph databases?



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# Homework

- Explain the concept of Data Modelling and the Entity-Relationship (ER) Model. Discuss entities, entity sets, attributes, keys, relationships, relationship degree, cardinality, and participation constraints with suitable examples. Also explain the role of ER modelling in designing a real-world database.





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Thanks!

📌 Heads Up!

These slides are just a **guide** — not the full story!

To really learn , make sure to read your book, and explore other resources too.